

ASSIGNMENT #01

Course: CHE 341: CHEMICAL PROCESS CONTROL
Term: Spring 2026
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Due Date: Friday, January 23rd TA's Mailbox by 3:00PM

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name(s) in the report.
- Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

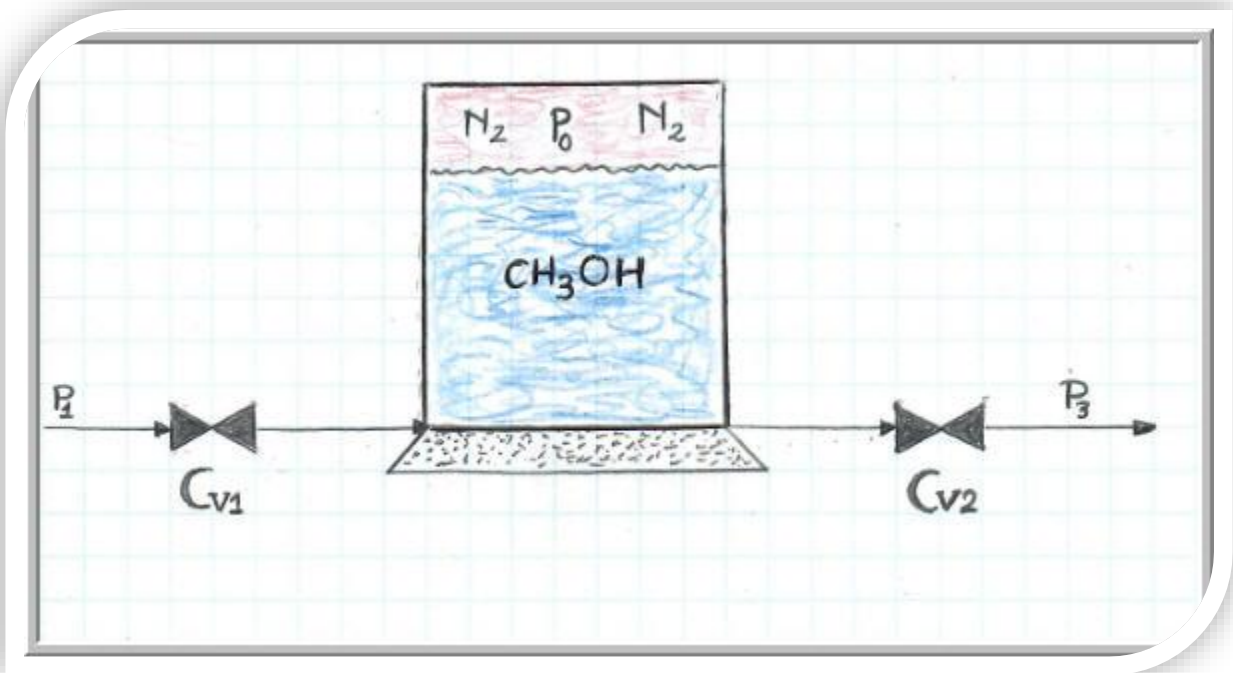
Problem A: Modeling of a Blanketed Storage Tank for Methanol

A **well-insulated** cylindrical tank (see Figure below) is used for storage of liquid methanol. The tank has volume of $V = 5000 \text{ ft}^3$ and the surface area of the bottom is $A = 200 \text{ ft}^2$. The liquid methanol is blanketed with nitrogen to avoid contamination with air and moisture. The feed and effluent line valve coefficients are $C_{v1} = C_{v2} = 110 \text{ gpm} / \sqrt{\text{psi}}$. The valves exhibit linear characteristics and the flow rate through the valves is given by the equations

$$F_1 = C_{v1} \sqrt{\frac{\Delta P_1}{s_g}} \quad \text{and} \quad F_2 = C_{v2} \sqrt{\frac{\Delta P_2}{s_g}}$$

where s_g is the specific gravity of methanol. The gauge pressure in the tank (measured in the nitrogen space) is $P_0 = 5 \text{ psig}$ and the temperature $T_0 = 25^\circ \text{C}$. The upstream to the feed line valve pressure is $P_1 = 33.5 \text{ psia}$ and the downstream to the effluent line valve pressure is $P_3 = 20.5 \text{ psia}$.

Figure: Methanol Storage Tank



Develop the necessary equation(s) required to monitor the liquid methanol height and the temperature of the vapor space as a function of time, due to various pressure disturbances in the feed and effluent lines. After you develop the model perform the following tasks

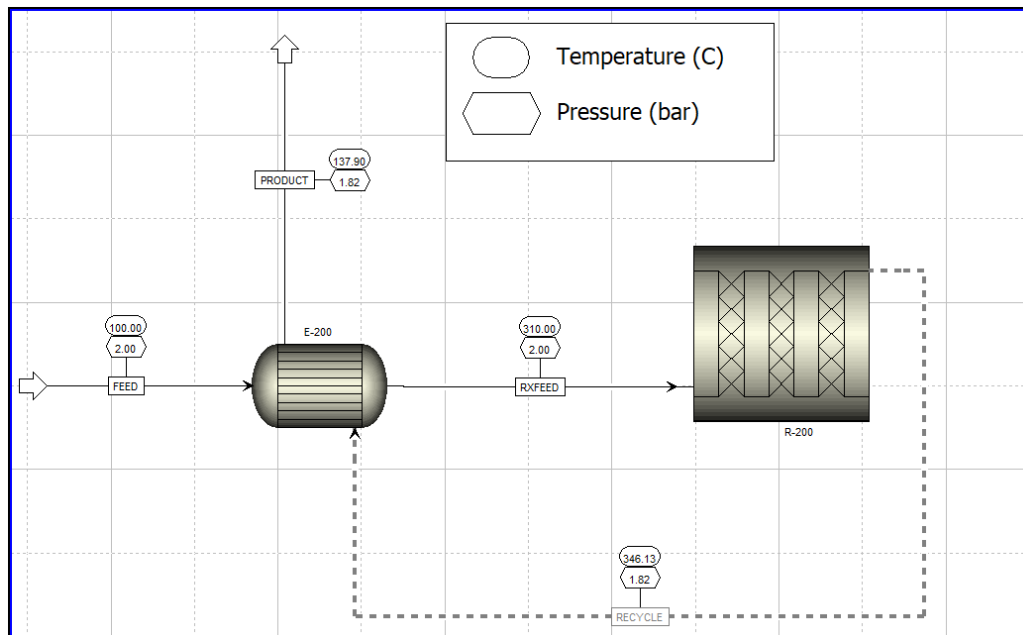
1. Calculate the steady-state liquid methanol height in **ft**
2. Calculate the number of moles of methanol and nitrogen in the vapor space
3. Perform a dynamic simulation for about 4 hr to demonstrate that the liquid height and temperature remain constant at the steady-state values in the absence of any disturbances
4. Start the simulation again at $t=0$ min. At time $t=2$ hr, the upstream to the feed valve pressure increases by 20%. Plot the liquid height and vapor space temperature as a function of time and calculate the new steady-state values.
5. Calculate the final pressure in **psia** of the vapor space in the storage tank
6. Calculate the time, in **min**, required to reach the new steady-state values defined as 99.5% of the asymptotic (steady-state) values.
7. In one Appendix of your report include all the plots of your analysis.
8. In a second Appendix of your report include the computer code with all the data that you used to do the numerical integration. You are free to use any software (like Matlab) but I would strongly recommend that you use the CHE Calculator

Data and Assumptions:

1. You may get the methanol and nitrogen properties from the ASPEN Plus simulator or the CHE Calculator (I would recommend the CHE Calculator)
2. Assume that nitrogen/methanol mixture in the vapor space behaves as an ideal gas with constant heat capacities C_p and C_v .
3. Assume that the amount of methanol in the vapor space does not change appreciably because of temperature and pressure variations.

Problem B: Design vs. Operational Models: Understanding Positive Feedback

Consider the ASPEN Plus simulation file that is uploaded on the UIC Blackboard along with your first assignment. The simulation file describes the catalytic oxidation of a hydrocarbon in an adiabatic fixed bed reactor. The feed to the reactor is preheated to a temperature of 310 °C using a CFE (Combined Feed Exchanger) as shown in the diagram below:



Perform the following tasks:

1. Study the simulation file and provide a description (no longer than one page) of what this design is all about with some detail on the reaction mechanism and the reactor design.
2. Modify the simulation file to reflect operational mode. The current one is the design mode.
3. Determine what will happen if due to some unforeseen disturbances the reactor effluent temperature increases by 2 °C. When answering this question assume that there are no controls in place. Describe in detail your answer and corroborate it with the proper simulations. Assuming that it takes 20 seconds for the disturbance to go through the process, plot the temperature of the reactor effluent versus time for the first 60 seconds after the onset of the disturbance
4. Recommend a solution to remedy the situation resulting from the unexpected rise in temperature and modify the simulation file to reflect your recommendation. Describe in detail your recommendation and report the results of your simulation that reflect your recommendation. Also calculate the net increase or decrease in utilities after the onset of the disturbance.

Solution - Problem A: Modeling of a Blanketed Storage Tank for Methanol

This analysis concerns the dynamic behavior of a cylindrical methanol storage tank that is blanketed with nitrogen to prevent contamination with moisture and air. The tank exchanges liquid methanol with the surroundings through two control valves: a feed valve and an effluent valve. Changes in upstream or downstream pressures affect the liquid level within the tank, as well as the temperature and pressure of the vapor space due to gas compression and expansion. The objective of this study is to determine the steady-state operating conditions, develop a dynamic model for the coupled liquid–gas system, simulate the system response to disturbances, and quantify key dynamic performance characteristics. Throughout the analysis, methanol and nitrogen thermophysical data are obtained from the CHE Calculator, which was created by Dr Michael Carcoitas, and the nitrogen/methanol vapor mixture is assumed to behave as an ideal gas with constant heat capacities.

1. Calculate the steady-state liquid methanol height

At steady state the inlet and outlet volumetric flowrates of liquid methanol are equal, so the two control valves (which have the same coefficient C_v) must have the same pressure drop:

$$F_1 = C_v \sqrt{\frac{\Delta P_1}{s_g}}, F_2 = C_v \sqrt{\frac{\Delta P_2}{s_g}}$$
$$F_1 = F_2 \Rightarrow \Delta P_1 = \Delta P_2$$

Let P_b be the pressure at the liquid connection on the tank bottom. Then

$$\Delta P_1 = P_1 - P_b, \quad \Delta P_2 = P_b - P_3$$

with $P_1 = 33.5$ psia (upstream of feed valve) and $P_3 = 20.5$ psia (downstream of effluent valve). Setting $\Delta P_1 = \Delta P_2$ gives

$$P_1 - P_b = P_b - P_3 \Rightarrow P_b = \frac{P_1 + P_3}{2} = \frac{33.5 + 20.5}{2} = 27.0 \text{ psia}$$

The vapor space above the liquid is blanketed with nitrogen at a gauge pressure of $P_0 = 5$ psig. In absolute units this is

$$P_{\text{vapor}} = 14.7 + 5.0 = 19.7 \text{ psia}$$

Therefore the hydrostatic pressure contribution of the methanol column is

$$\Delta P_h = P_b - P_{\text{vapor}} = 27.0 - 19.7 = 7.3 \text{ psi}$$

To convert this pressure head into a liquid height we need the specific gravity of methanol. From the property table:

- Molecular weight of methanol: $M = 32.042$ kg/kmol

- Liquid molar volume at NBP: $V_m = 39.73 \text{ cm}^3/\text{mol}$

The density is:

$$\rho = \frac{M}{V_m} = \frac{32.042 \text{ g/mol}}{39.73 \text{ cm}^3/\text{mol}} \approx 0.806 \text{ g/cm}^3$$

Thus, the specific gravity (relative to water at 1.0 g/cm^3) is

$$s_g = 0.806$$

Using the standard relation $1 \text{ psi} = 2.31 \text{ ft of water}$, the height of a liquid with specific gravity s_g that produces a pressure drop ΔP_h is

$$\Delta P_h = \frac{s_g h}{2.31} \Rightarrow h = \frac{\Delta P_h 2.31}{s_g}$$

$$h = \frac{7.3 \times 2.31}{0.806} \approx 20.92 \text{ ft}$$

$$h_{\text{steady}} = \mathbf{20.92 \text{ ft}} \text{ of liquid methanol}$$

2. Calculate the number of moles of methanol and nitrogen in the vapor space

The moles of vapor-phase components are determined by first calculating the vapor-space volume and then applying the ideal-gas law using the partial pressures of methanol and nitrogen.

2.1 Vapor-space volume

The total tank volume and cross-sectional area are:

$$V_{\text{tot}} = 5000 \text{ ft}^3, \quad A = 200 \text{ ft}^2$$

so the internal height is:

$$H_{\text{tank}} = \frac{V_{\text{tot}}}{A} = \frac{5000}{200} = 25 \text{ ft}$$

From part 1, the steady-state liquid height is:

$$h_{\text{steady}} = 20.92 \text{ ft}$$

Thus, the vapor height and vapor-space volume are:

$$h_v = H_{\text{tank}} - h_{\text{steady}} = 25 - 20.92 = 4.08 \text{ ft}$$

$$V_v = A h_v = 200 \times 4.08 = 816 \text{ ft}^3$$

2.2 Total moles of gas in the vapor space

The vapor space is blanketed with nitrogen at 5 psig. In absolute units:

$$P_{\text{vapor}} = P_{\text{atm}} + P_{\text{gauge}} = 14.7 + 5.0 = 19.7 \text{ psia}$$

The system temperature:

$$T_0 = 25^\circ\text{C} = 77^\circ\text{F} = 537^\circ\text{R}$$

Using the ideal-gas law with $R = 10.731 \text{ ft}^3 \text{ psia}/(\text{lbmol} \cdot ^\circ\text{R})$:

$$n_{\text{total}} = \frac{P_{\text{vapor}} V_v}{R T_0} = \frac{(19.7)(816)}{(10.731)(537)} \approx \mathbf{2.79 \text{ lbmol}}$$

2.3 Distribution between methanol and nitrogen

Since Henry's constant for methanol and nitrogen is close to 1, partial pressures can be used to find their respective vapor pressures. At 25°C , the saturation vapor pressure of methanol obtained from the CHE Calculator is:

$$P_{\text{sat, MeOH}}(25^\circ\text{C}) = 0.15868 \text{ bar}$$

Converting to psia using $1 \text{ bar} = 14.5038 \text{ psia}$:

$$P_{\text{MeOH}} = 0.15868 \times 14.5038 \approx 2.30 \text{ psia}$$

Assuming vapor-liquid equilibrium, the methanol partial pressure equals its saturation pressure. The remainder of the total pressure is nitrogen:

$$P_{N_2} = P_{\text{vapor}} - P_{\text{MeOH}} = 19.7 - 2.30 = 17.4 \text{ psia}$$

Applying the ideal-gas law to each component:

$$\begin{aligned} n_{\text{MeOH}} &= \frac{P_{\text{MeOH}} V_v}{R T_0} = \frac{(2.30)(816)}{(10.731)(537)} \approx \mathbf{0.326 \text{ lbmol}} \\ n_{N_2} &= \frac{P_{N_2} V_v}{R T_0} = \frac{(17.4)(816)}{(10.731)(537)} \approx \mathbf{2.46 \text{ lbmol}} \end{aligned}$$

Summing:

$$n_{\text{MeOH}} + n_{N_2} \approx 0.326 + 2.46 = 2.786 \text{ lbmol}$$

which matches the total previously calculated ($\approx 2.79 \text{ lbmol}$), confirming consistency.

3. Dynamic simulation for 4 hr with no disturbances

In this section the objective is to verify that, when started at the steady-state conditions and with all boundary pressures held constant, the tank level and vapor temperature remain essentially constant in time. This is done by deriving a dynamic model, implementing it in the CHE Calculator ODE integrator through the user function FODEA1, and simulating over a 4-hr period. The plots and tables generated by the integrator are included in the Appendix and are referenced below.

3.1 Choice of state variables

Two coupled state variables are sufficient to describe the dynamics:

- $h(t)$: liquid methanol height in the tank (ft)
- $T(t)$: vapor-space temperature ($^{\circ}\text{R}$)

The vapor is treated as a closed ideal-gas mixture with constant total moles n_g (sum of methanol and nitrogen moles from Question 2). The composition is assumed constant in time, in accordance with the problem statement.

In the ODE function FODEA1, these states are stored as:

- $u(1) = h$
- $u(2) = T$

The function returns the derivatives dh/dt and dT/dt as a 2-element array.

3.2 Liquid mass balance and valve flows

The total volumetric flow into and out of the tank is determined by the two control valves. For each valve the CHE Calculator gives a linear characteristic:

$$F = C_v \sqrt{\frac{\Delta P}{s_g}}$$

where

- F = volumetric flowrate of liquid methanol (ft^3/hr),
- C_v = valve coefficient converted to $\text{ft}^3/\text{hr}/\sqrt{\text{psi}}$,
- ΔP = pressure drop across the valve (psi),
- s_g = specific gravity of methanol (= 0.806).

For the **feed valve**:

$$F_1 = C_v^* \sqrt{\frac{\Delta P_1}{s_g}}, \Delta P_1 = P_1 - P_b$$

For the **effluent valve**:

$$F_2 = C_v^* \sqrt{\frac{\Delta P_2}{s_g}}, \Delta P_2 = P_b - P_3$$

with $P_1 = 33.5$ psia and $P_3 = 20.5$ psia fixed in this section.

C_v^* is the valve coefficient converted from gpm/ $\sqrt{\text{psi}}$ to ft³/hr/ $\sqrt{\text{psi}}$ and is listed in the parameter table on the spreadsheet.

The bottom pressure P_b is the sum of the vapor pressure and the hydrostatic head of the liquid column:

$$P_b = P_g + \Delta P_h = P_g + \frac{s_g h}{2.31}$$

where P_g is the gas pressure (psia) and the factor 2.31 ft/psi converts from pressure to feet of water.

A dynamic mass balance on the liquid gives:

$$\frac{dV_L}{dt} = F_1 - F_2$$

Since $V_L = Ah$ and A is the constant cross-sectional area of the tank,

$$\frac{dh}{dt} = \frac{F_1 - F_2}{A}$$

This equation is coded in FODEA1 to compute dh/dt.

3.3 Gas-phase volume, pressure, and energy balance

The vapor volume is the remainder of the tank:

$$V_v = V_{\text{tot}} - V_L = V_{\text{tot}} - Ah$$

The gas pressure follows the ideal gas law with constant moles n_g :

$$P_g = \frac{n_g RT}{V_v}$$

This relation is evaluated inside FODEA1 at each time step from the current values of h and T .

Because the tank is well insulated and no mass of gas enters or leaves, the gas behaves as a closed, adiabatic system. For an ideal gas with total heat capacity $n_g C_v$, the first law gives:

$$n_g C_v \frac{dT}{dt} = -P_g \frac{dV_v}{dt}$$

Using $V_v = V_{\text{tot}} - Ah$, we have:

$$\frac{dV_v}{dt} = -A \frac{dh}{dt}$$

Substituting:

$$n_g C_v \frac{dT}{dt} = -P_g (-A) \frac{dh}{dt} = P_g A \frac{dh}{dt}$$

It is convenient to express this in terms of the ratio $\gamma = C_p/C_v$. For an ideal gas, $C_p - C_v = R$, which leads (after some algebra) to the familiar form:

$$\frac{dT}{dt} = -(\gamma - 1) \frac{T}{V_v} \frac{dV_v}{dt}$$

and since $\frac{dV_v}{dt} = -A \frac{dh}{dt}$, this becomes

$$\frac{dT}{dt} = (\gamma - 1) \frac{T}{V_v} A \frac{dh}{dt}$$

In **FODEA1**, we first compute dh/dt from the liquid balance, then use the above expression to compute dT/dt . The constant γ (taken as 1.4 for the nitrogen-rich mixture) and the total moles n_g are passed to the function as parameters from the spreadsheet.

3.4 Implementation in CHE Calculator (FODEA1 and RungeKutta5)

The dynamic model was implemented in the CHE Calculator using the user-defined ODE function FODEA1 together with the built-in RungeKutta5 integrator and the ODE Setup utility.

First, the ODE Setup tool was used to specify:

- the beginning and end of integration (0 to 4 hr)
- the total number of solution points and the number of steps between solution points
- the relative and absolute state tolerances

ODE Setup then writes these values into the *ODE Integrator Options* block and creates a *Problem Data* block where all physical parameters (tank area, total volume, valve coefficient, specific gravity, upstream and downstream pressures, total gas moles, gas constant, and heat-

capacity ratio) are entered. This Problem Data block is later passed to the ODE function as the parameter range.

The actual time integration is performed by the worksheet function RungeKutta5, which is entered in the first solution row of the table. In CHE Calculator its arguments are:

- the name of the ODE function (“FODEA1”)
- the previous time value t_n
- the new time value t_{n+1}
- the vector of previous state values u_n (liquid height and vapor temperature)
- the number of internal Runge–Kutta steps between solution points
- the range containing the physical parameter data

For each row, RungeKutta5 takes the current interval $[t_n, t_{n+1}]$ and the current state vector, calls FODEA1 multiple times within that interval to evaluate the right-hand sides of the ODEs, and returns the updated state vector u_{n+1} . FODEA1 itself does not perform any integration; it only computes the instantaneous derivatives dh/dt and dT/dt from the model equations using the supplied time, state values, and parameters. The specific worksheet formula and arguments are provided in Appendix 2, and the resulting tables and plots are included in Appendix 1.

The user-defined function FODEA1 contains the mathematical model that defines the system dynamics. When RungeKutta5 calls it, FODEA1 reads the physical parameters, computes the vapor volume and pressure from the ideal gas law, calculates the bottom pressure and valve pressure drops, determines the inlet and outlet flowrates, and then evaluates the liquid mass balance to obtain dh/dt and the adiabatic energy balance to obtain dT/dt . These two derivatives are returned to RungeKutta5, which performs the numerical integration; the RungeKutta5 formula is then filled down the column to generate the full time history. The full VBA implementation of FODEA1 is provided in Appendix 2.

3.5 Results and interpretation

The simulation results (time series table and plots of $h(t)$ and $T(t)$) are presented in the Appendix. Both variables remain essentially constant over the 4-hr period:

- $h(t)$ stays at 20.92 ft within less than 10^{-3} ft.
- $T(t)$ remains at approximately 537 °R with negligible drift.

These tiny variations are due only to numerical integration error and rounding of physical parameters; physically, the system is at steady state and exhibits no dynamic response in the absence of disturbances. This confirms that the derived model and initial conditions are consistent with the specified steady-state operation.

4. Dynamic simulation with a 20% increase in upstream feed pressure

In this section, the effect of a disturbance on the tank dynamics was examined. Starting from the steady-state initial conditions established in Question 3, the upstream pressure to the feed valve (P_1) was increased by 20% at $t = 2$ hr, while all other boundary conditions and physical assumptions remained unchanged. The simulation demonstrates how the liquid height and vapor temperature respond to this disturbance and settle to a new steady-state value. The resulting plots

and time-series data are provided in Appendix 1, and the corresponding spreadsheet setup, code snippets, and run configurations are shown in Appendix 2.

4.1 Differences from the previous simulation

The mathematical formulation of the ODEs did not change between Questions 3 and 4. The same liquid-level and energy balances apply, and the same state variables (h, T) were integrated in time. The only physical change introduced was the modification of the feed-valve driving pressure after 2 hours, which alters the inlet flowrate and therefore the tank dynamics. Specifically:

$$P_1(t) = \begin{cases} 33.5 \text{ psia}, & t < 2 \text{ hr} \\ 1.20 \times 33.5 = 40.2 \text{ psia}, & t \geq 2 \text{ hr} \end{cases}$$

This increased upstream pressure produces a higher inlet flowrate through the feed valve, causing the level to rise and compressing the vapor space, which increases both the pressure and temperature of the gas. A new steady state is reached once inflow again matches outflow.

4.2 Modifications to the ODE function (FODEA2)

To implement this disturbance, a second ODE function named FODEA2 was created. FODEA2 is identical to FODEA1 in terms of the model equations and returned derivatives, except that the upstream pressure P_1 is made time-dependent. Rather than using a constant value from the parameter block, P_1 is now defined internally using an If/Then-style conditional based on the current time value passed by RungeKutta5. In descriptive terms, FODEA2 performs the following additional step:

- Before computing valve flows, check the current integration time.
 - If $t < 2\text{hr}$, use the original P_1 value from the parameter block.
 - If $t \geq 2\text{hr}$, multiply that parameter value by 1.20.

This single modification ensures that all subsequent computations (bottom pressure, valve drops, flowrates, dh/dt , and dT/dt) automatically reflect the disturbance. The full VBA listing of FODEA2 and screenshots of the corresponding changes in the spreadsheet are provided in Appendix 2.

4.3 Changes to the spreadsheet setup

The ODE Setup utility was reused without changes, and the RungeKutta5 call remained structurally identical to that used in Question 3. The only difference is that the name of the ODE function provided to RungeKutta5 was changed from "FODEA1" to "FODEA2". All other inputs—including the initial conditions, number of equations, number of internal steps, and parameter range—remained the same. This substitution allowed CHE Calculator to reuse the same integration framework while incorporating the new time-dependent boundary condition.

4.4 Observed system response

The simulation was run from $t = 0$ to $t = 4$ hr to capture both the transient response and the relaxation to the new steady state. As shown in the plots in Appendix 1, both $h(t)$ and $T(t)$ remain constant until $t = 2$ hr, at which point they rise smoothly toward new steady-state values. After sufficient time, the liquid height settled at approximately **21.34 ft**, and the vapor temperature settled at approximately **560.6 °R**, demonstrating the expected physical behavior under a sustained increase in inlet pressure.

5. Final vapor-space pressure after the disturbance

After the 20% increase in upstream feed pressure at $t = 2$ hr, both the liquid level and vapor temperature increased and eventually reached new steady-state values. Because no gas enters or leaves the vapor space, the total number of moles in the gas phase remains constant, and the final pressure in the vapor space can be determined directly from the ideal gas law once the final temperature and vapor volume are known. The relevant dynamic results (final h and T) are taken from the simulation output provided in Appendix 1.

5.1 Determining the final vapor-space volume

From the simulation results, the final steady-state liquid level after the disturbance was approximately:

$$h_{\text{final}} \approx 21.34 \text{ ft}$$

The vapor-space volume is given by:

$$V_v = V_{\text{tot}} - Ah$$

Using:

- total tank volume: $V_{\text{tot}} = 5000 \text{ ft}^3$
- tank area: $A = 200 \text{ ft}^2$

gives:

$$V_{v,\text{final}} = 5000 - 200(21.34) \approx 732.8 \text{ ft}^3$$

5.2 Applying the ideal gas law

From earlier calculations (Question 2), the total moles of gas in the vapor space remain constant at:

$$n_g \approx 2.79 \text{ lbmol}$$

The final vapor temperature from the simulation was:

$$T_{\text{final}} \approx 560.6 \text{ °R}$$

Applying the ideal gas law:

$$P_{\text{final}} = \frac{n_g R T_{\text{final}}}{V_{v,\text{final}}}$$

with $R = 10.731 \text{ ft}^3 \text{ psia}/(\text{lbmol} \cdot ^\circ \text{R})$, gives:

$$P_{\text{final}} = \frac{(2.79)(10.731)(560.6)}{732.8} \approx \mathbf{22.9 \text{ psia}}$$

This value represents the new steady-state vapor pressure in the tank after the inlet-pressure disturbance. It is higher than the initial value (19.7 psia), as expected, due to compression and heating of the vapor space caused by the increased liquid level and temperature.

6. Time required to reach 99.5% of the new steady-state values

The final part of the analysis is to quantify how quickly the system responds to the disturbance applied in Question 4. Specifically, the required settling time is defined as the time needed for both the liquid height and vapor temperature to reach 99.5% of their asymptotic (steady-state) values following the 20% step increase in upstream feed pressure at $t = 2 \text{ hr}$.

6.1 Definition of the 99.5% criterion

For each state variable $x(t)$ (either $h(t)$ or $T(t)$), the 99.5% target value is defined as:

$$x_{99.5} = x_0 + 0.995 (x_{\text{ss}} - x_0)$$

where:

- x_0 is the initial value just **before** the disturbance (the original steady state),
- x_{ss} is the new steady-state value obtained **after** the disturbance, and
- $x_{99.5}$ is the value corresponding to 99.5% of the total change.

From the simulation in Appendix 1:

- Initial liquid level: $h_0 \approx 20.92 \text{ ft}$
- Final liquid level: $h_{\text{ss}} \approx 21.34 \text{ ft}$

so:

$$h_{99.5} = 20.92 + 0.995 (21.34 - 20.92) \approx 21.33 \text{ ft}$$

Similarly, for temperature:

- Initial temperature: $T_0 \approx 537 \text{ }^\circ \text{R}$
- Final temperature: $T_{\text{ss}} \approx 560.6 \text{ }^\circ \text{R}$

so:

$$T_{99.5} = 537 + 0.995 (560.6 - 537) \approx 560.5 \text{ }^{\circ}\text{R}$$

These values represent the thresholds used to determine when the system has effectively reached the new steady state.

6.2 Use of simulation data to determine settling time

The time-series data for $h(t)$ and $T(t)$ generated by the RungeKutta5 integrator (see Appendix 1) were examined to find the first time at which each variable exceeded its respective 99.5% target:

- For the liquid height, the table shows that $h(t)$ crosses $h_{99.5} \approx 21.33$ ft shortly after the disturbance and then remains above this value without overshoot.
- For the vapor temperature, $T(t)$ crosses $T_{99.5} \approx 560.5 \text{ }^{\circ}\text{R}$ slightly later than the level, making temperature the slower of the two states.

To estimate the time more accurately, the two solution points in the table that bracket the 99.5% value were identified for each variable, and a simple linear interpolation was applied between them. This procedure gives approximately the same settling time for both variables, with temperature controlling.

From this analysis, the time at which $T(t)$ first satisfies $T(t) \geq T_{99.5}$ is approximately:

$$t_{99.5} \approx 2.45 \text{ hr}$$

Since the disturbance was applied at $t = 2.0$ hr, the settling time after the step change is:

$$t_{\text{settle}} = t_{99.5} - 2.0 \approx 0.45 \text{ hr} \approx \mathbf{27 \text{ min}}$$

thus, Based on the 99.5% criteria and the simulation data, the tank requires approximately 27 minutes after the disturbance to reach the new steady state. This settling time reflects the combined dynamics of level and gas compression, with the vapor temperature providing the limiting (slowest) response.

7. Appendix I: Plots and simulation results

Appendix 1 presents the tabulated simulation results and corresponding plots of liquid height and vapor temperature as functions of time for both the undisturbed case and the disturbed case examined in Questions 3–6.

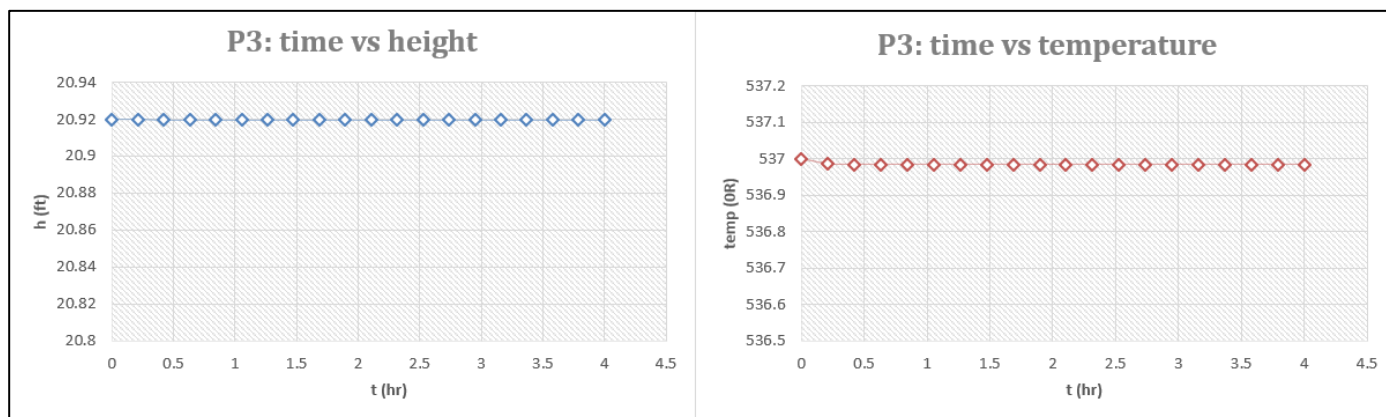


Figure 1: part 3- height & temperature vs time graphs

	A	B	C	D	E	F	G	H	I
1	ODE Integrator Options					Time	h(ft)	T °R	
2	Beginning of Integration	0				0	20.92	537	
3	End of Integration	4				0.210526	20.91971	536.9847	
4	Total Number of Solution Points	20				0.421053	20.91969	536.9839	
5	Intermediate Solution Breakpoints					0.631579	20.91969	536.9838	
6	Number of Steps between Solution Points	10				0.842105	20.91969	536.9838	
7	User Calculated Integration Step Size	0.021053				1.052632	20.91969	536.9838	
8						1.263158	20.91969	536.9838	
9	Number of Equations	2				1.473684	20.91969	536.9838	
10	Relative State Tolerance	0.000001				1.684211	20.91969	536.9838	
11	Absolute State Tolerance	0.000001				1.894737	20.91969	536.9838	
12	Estimated Local Truncation Error					2.105263	20.91969	536.9838	
13	Estimated Initial Integration Step Size					2.315789	20.91969	536.9838	
14						2.526316	20.91969	536.9838	
15	PROBLEM Data					2.736842	20.91969	536.9838	
16	A, ft^2	200				2.947368	20.91969	536.9838	
17	Vtot, ft^3	5000				3.157895	20.91969	536.9838	
18	Cv*, ft^3/hr/sqrt(psi)	882.295				3.368421	20.91969	536.9838	
19	sg	0.806				3.578947	20.91969	536.9838	
20	P1, psia	33.5				3.789474	20.91969	536.9838	
21	P3, psia	20.5				4	20.91969	536.9838	
22	n_g, lbmol	2.79							
23	R, ft^3.psia/(lbmol*R)	10.731							
24	gamma	1.4							
25									

Figure 2: part 3- height & temperature vs time ODE results

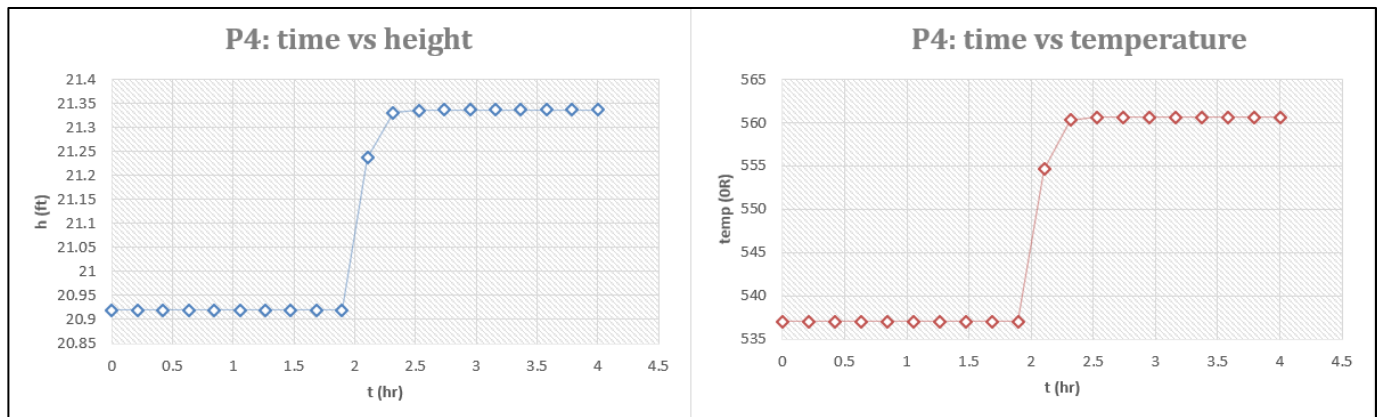


Figure 3: part 4- height & temperature vs time graphs

	A	B	C	D	E	F	G	H	I
1	ODE Integrator Options					Time	h(ft)	T @	
2	Beginning of Integration	0				0	20.92	537	
3	End of Integration	4				0.210526	20.91971	536.9847	
4	Total Number of Solution Points	20				0.421053	20.91969	536.9839	
5	Intermediate Solution Breakpoints					0.631579	20.91969	536.9838	
6	Number of Steps between Solution Points	10				0.842105	20.91969	536.9838	
7	User Calculated Integration Step Size	0.021053				1.052632	20.91969	536.9838	
8						1.263158	20.91969	536.9838	
9	Number of Equations	2				1.473684	20.91969	536.9838	
10	Relative State Tolerance	0.000001				1.684211	20.91969	536.9838	
11	Absolute State Tolerance	0.000001				1.894737	20.91969	536.9838	
12	Estimated Local Truncation Error					2.105263	21.23666	554.6362	
13	Estimated Initial Integration Step Size					2.315789	21.33104	560.3003	
14						2.526316	21.33589	560.5966	
15	PROBLEM Data					2.736842	21.33613	560.6114	
16	A, ft ²	200				2.947368	21.33614	560.6121	
17	Vtot, ft ³	5000				3.157895	21.33615	560.6121	
18	Cv*, ft ³ /hr/sqrt(psi)	882.295				3.368421	21.33615	560.6121	
19	sg	0.806				3.578947	21.33615	560.6121	
20	P1, psia	33.5				3.789474	21.33615	560.6121	
21	P3, psia	20.5				4	21.33615	560.6121	
22	n_g, lbmol	2.79							
23	R, ft ³ .psia/(lbmol*R)	10.731							
24	gamma	1.4							
25									

Figure 4: part 4- height & temperature vs time ODE results

8. Appendix II: VBA code

Appendix 2 presents the full VBA code for FODEA1 and FODEA2, which were used to solve Questions 3–6.

```
Function fODEA1(T As Double, u() As Double, Optional Param As Range) As Double()  
    ' State variables  
    Dim h As Double, Tv As Double  
    h = u(1)          ' liquid height, ft  
    Tv = u(2)         ' vapor temperature, °R  
  
    ' Parameters from worksheet  
    Dim A As Double, Vtot As Double, CvStar As Double, sg As Double  
    Dim P1 As Double, P3 As Double, nGas As Double, Rgas As Double, gamma As Double  
  
    A = Param.Cells(1, 1)      ' 200 ft^2  
    Vtot = Param.Cells(2, 1)   ' 5000 ft^3  
    CvStar = Param.Cells(3, 1) ' 882.3 ft^3/hr/sqrt(psi)  
    sg = Param.Cells(4, 1)     ' 0.806  
    P1 = Param.Cells(5, 1)     ' 33.5 psia  
    P3 = Param.Cells(6, 1)     ' 20.5 psia  
    nGas = Param.Cells(7, 1)   ' 2.79 lbmol  
    Rgas = Param.Cells(8, 1)   ' 10.731  
    gamma = Param.Cells(9, 1)  ' ~1.4  
  
    ' Local variables  
    Dim Vv As Double, Pgas As Double, Pbottom As Double  
    Dim dP1 As Double, dP2 As Double  
    Dim F1 As Double, F2 As Double  
    Dim dhdt As Double, dVvdt As Double, dTdt As Double  
    Dim f() As Double  
    Dim n As Integer  
  
    ' Vapor volume  
    Vv = Vtot - A * h  
  
    ' Gas pressure  
    Pgas = nGas * Rgas * Tv / Vv  
  
    ' Bottom pressure (include hydrostatic head)  
    Pbottom = Pgas + sg * h / 2.31  
  
    ' Valve pressure drops  
    dP1 = P1 - Pbottom  
    dP2 = Pbottom - P3  
    If dP1 < 0 Then dP1 = 0  
    If dP2 < 0 Then dP2 = 0  
  
    ' Valve flows (ft^3/hr)  
    F1 = CvStar * Sqr(dP1 / sg)  
    F2 = CvStar * Sqr(dP2 / sg)  
  
    ' ODEs  
    dhdt = (F1 - F2) / A  
    dVvdt = -A * dhdt  
  
    ' Return derivative vector  
    n = UBound(u)  
    ReDim f(1 To n) As Double  
    f(1) = dhdt  
    f(2) = dTdt  
    fODEA1 = f  
End Function
```

Figure 5: part 3- Full VBA code for FODEA1

```

Function FODEA2(T As Double, u() As Double, Optional Param As Range) As Double()
    Dim h As Double, Tv As Double
    h = u(1)          ' liquid height, ft
    Tv = u(2)         ' vapor temperature, °R

    ' Parameters from worksheet
    Dim A As Double, Vtot As Double, CvStar As Double, sg As Double
    Dim P1 As Double, P3 As Double, nGas As Double, Rgas As Double, gamma As Double

    A = Param.Cells(1, 1)
    Vtot = Param.Cells(2, 1)
    CvStar = Param.Cells(3, 1)
    sg = Param.Cells(4, 1)

    ' --- time-dependent disturbance in P1 ---
    If T < 2# Then
        P1 = Param.Cells(5, 1)          ' old P1 = 33.5 psia
    Else
        P1 = 1.2 * Param.Cells(5, 1)    ' +20% step to 40.2 psia
    End If
    ' -----

    P3 = Param.Cells(6, 1)
    nGas = Param.Cells(7, 1)
    Rgas = Param.Cells(8, 1)
    gamma = Param.Cells(9, 1)

    ' Local variables
    Dim Vv As Double, Pgas As Double, Pbottom As Double
    Dim dP1 As Double, dP2 As Double
    Dim F1 As Double, F2 As Double
    Dim dhdt As Double, dVvdt As Double, dTdt As Double
    Dim f() As Double
    Dim n As Integer

    ' Vapor volume
    Vv = Vtot - A * h

    ' Gas pressure
    Pgas = nGas * Rgas * Tv / Vv

    ' Bottom pressure (include hydrostatic head)
    Pbottom = Pgas + sg * h / 2.31

    ' Valve pressure drops
    dP1 = P1 - Pbottom
    dP2 = Pbottom - P3
    If dP1 < 0 Then dP1 = 0
    If dP2 < 0 Then dP2 = 0

    ' Valve flows (ft^3/hr)
    F1 = CvStar * Sqr(dP1 / sg)
    F2 = CvStar * Sqr(dP2 / sg)

    ' ODEs
    dhdt = (F1 - F2) / A
    dVvdt = -A * dhdt
    dTdt = -(gamma - 1#) * (Tv / Vv) * dVvdt

    ' Return derivative vector
    n = UBound(u)
    ReDim f(1 To n) As Double
    f(1) = dhdt
    f(2) = dTdt
    fODEA2 = f
End Function

```

Figure 6: part 4- Full VBA code for FODEA2

Solution - Problem B: Design Models vs. Operational Models

1. Process Description

The ASPEN Plus simulation models a continuous gas-phase catalytic oxidation of aromatic hydrocarbons to produce anhydride products. The feed is an air–hydrocarbon mixture containing oxygen, nitrogen, and aromatic species such as xylene isomers, with CO₂ and water included as possible products or inerts. Inside the reactor these aromatics are oxidized over a solid catalyst to form primarily maleic anhydride and phthalic anhydride, together with CO₂ and H₂O. The property method is SRK, which is appropriate for high-temperature, primarily vapor-phase systems containing light gases and moderately heavy organics, so the simulation can represent phase behavior and enthalpy changes reasonably well without introducing unnecessary complexity.

Upstream of the reactor, the feed passes through heat exchanger E-200, which is modeled as a shortcut exchanger in design mode. The exchanger specification is the cold-stream outlet temperature of the feed (310 °C); Aspen adjusts the required heat duty (and implied UA) to meet this target using the hot reactor effluent as the heating medium. This configuration is a combined feed–effluent exchanger: it recovers a portion of the heat released in the reactor to preheat the incoming feed, reducing the demand for external utilities and ensuring the reactor inlet is hot enough for the oxidation reactions to proceed at an appreciable rate. At the same time, because hotter reactor effluent leads to hotter feed, this unit creates a built-in positive feedback between reactor temperature and feed preheat, which is central to the “design vs. operational” discussion in the assignment.

After preheating, the stream enters R-200, which is modeled as an adiabatic plug-flow reactor (RPlug) with vapor-only phase and a multitubular packed-bed configuration. Catalyst loading and bed properties are specified so that reaction rates are evaluated on a per-catalyst basis, while pressure drop is neglected to keep the focus on thermal and compositional behavior. The reaction set R-1 represents a network of vapor-phase catalytic oxidation reactions. These include the selective oxidation of o-xylene to phthalic anhydride with water formation, competing partial oxidation routes that form maleic anhydride, and multiple over-oxidation pathways in which o-xylene or intermediate anhydrides are further converted to CO₂ and H₂O. All reactions are modeled using power-law kinetics with rates driven by reactant partial pressures in the gas phase, reflecting surface-controlled catalytic behavior without explicitly modeling adsorption. Because these oxidation reactions are strongly exothermic, the adiabatic assumption leads to a significant temperature rise along the reactor as oxygen is consumed and products form. This setup emphasizes how selectivity, conversion, and temperature are tightly coupled, and why the reactor is inherently sensitive to changes in feed composition or inlet temperature.

Taken together, the flowsheet represents a tightly coupled reactor–heat-recovery system: the exothermic reactor generates heat, the CFE recycles part of that heat into the feed, and the recycle structure allows composition and temperature changes to propagate around the loop. The simulation is therefore not just a collection of unit operations, but a testbed for understanding how design choices interact to determine overall process stability and responsiveness in the absence of active control.

2. Operational mode modification

To transition the flowsheet from design mode to operational mode, the heat exchanger E-200 was treated as the key piece of equipment whose behavior changes between these two perspectives. First, the exchanger was solved in design mode, where the cold-stream outlet temperature is specified and Aspen calculates the required heat-transfer performance. From this completed design case, the resulting overall heat-transfer parameter (UA) was obtained from the exchanger results. This UA value represents a physically sized piece of equipment and therefore reflects a realistic operating constraint.

The exchanger was then switched to simulation (off-design) mode, and the previously calculated UA was fixed as an input. With UA locked, the heat exchanger no longer forces the feed to reach a target outlet temperature. Instead, the actual feed temperature becomes an output that depends on the reactor effluent temperature and the available heat-transfer capacity. This change converts the exchanger from a design tool into an operational unit and allows natural process feedback between the reactor and the feed preheat to emerge. The operational mode heat exchanger specification is shown in figure 7.

The screenshot shows the 'E-200 (HeatX)' specification window in Aspen Plus. The 'Specifications' tab is active. Under 'Model fidelity', 'Shortcut' is selected. Under 'Hot fluid', 'Shell' is selected. Under 'Shortcut flow direction', 'Countercurrent' is selected. The 'Calculation mode' is set to 'Simulation'. In the 'Exchanger specification' section, 'Cold stream outlet temperature' is specified with a value of 310 C. The 'Exchanger area' is 646.735 sqm, and the 'Constant UA' is 646.735 kW/K. The 'Minimum temperature approach' is 1 C. Buttons for 'Size Exchanger', 'Specify Geometry', and 'Results' are at the bottom.

Specification	Value	Units
Cold stream outlet temperature	310	C
Exchanger area	646.735	sqm
Constant UA	646.735	kW/K
Minimum temperature approach	1	C

Figure 7: Operational Mode Heat Exchanger

No changes were made to the reactor model during this transition. The reactor already represents normal operation: it has fixed geometry and catalyst loading, uses specified kinetic rate expressions, and operates adiabatically without pressure drop. These features correspond to physical constraints that would not change between design and operation. As a result, locking the heat exchanger UA is sufficient to shift the overall flowsheet into an operational framework.

3. Response to an Uncontrolled Temperature Disturbance

To study the effect of an unexpected temperature increase in the reactor effluent, a disturbance was introduced by increasing the temperature of the recycle (tear) stream by 2 °C, which is adjusted from . The recycle stream represents the reactor effluent returning to the combined feed–effluent heat exchanger (E-200), so increasing its temperature directly simulates a sudden rise in reactor outlet temperature. This disturbed recycle temperature serves as the initial condition for the transient analysis and represents the onset of the disturbance.

The heat exchanger was operated in simulation mode with a fixed UA rather than a fixed outlet temperature. As a result, the feed temperature to the reactor is no longer controlled and instead responds to changes in the hot-side temperature. After the disturbance, the hotter recycle stream transfers more heat to the incoming feed, increasing the reactor inlet temperature. Because the reactor is adiabatic and the oxidation reactions are strongly exothermic, this higher inlet temperature increases reaction rates and heat release, producing an even hotter reactor effluent. That hotter effluent then returns to the heat exchanger, further increasing feed preheat. This sequence creates a positive feedback loop between the reactor and the heat exchanger.

The system response was evaluated in discrete steps, with each full pass through the heat exchanger and reactor representing 20 seconds of real time. The time axis for the transient plot begins at $t = 0$ s, which corresponds to the moment immediately after the disturbance is applied (i.e., after the recycle temperature is increased). The original steady-state condition prior to the disturbance is not plotted. Starting the plot at the disturbed condition isolates the dynamic response to the disturbance itself. If the plot had instead included the pre-disturbance steady state or an artificial zero-temperature reference, the numerical values would differ, but the same runaway trend would still be observed.

As shown in the attached table and figure, the reactor effluent temperature increases rapidly with each loop of the system. Within a few iterations, the temperature rises sharply, and the simulation eventually produces warnings and convergence errors, indicating thermal runaway. This behavior demonstrates that, in the absence of controls, a fixed-UA feed–effluent exchanger coupled with an adiabatic exothermic reactor can be inherently unstable when subjected to even small temperature disturbances.

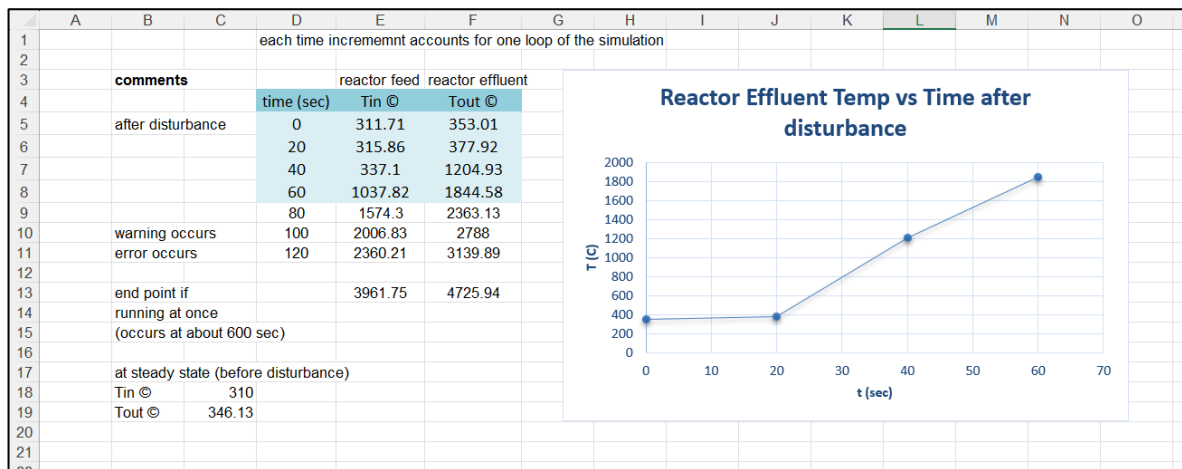


Figure 8: Reactor Effluent Temperature after Disturbance

4. Temperature Runaway Solution

To prevent the temperature runaway in operational mode, I added a heater (E-201) between the combined feed–effluent exchanger and the reactor and specified a fixed reactor inlet temperature of 310 °C. This keeps the reactor feed at the same target used in the original design case, so the reactor sees the intended inlet condition even when the upstream exchanger is switched to simulation mode (fixed UA). I also set no pressure drop in this added heater so it only affects temperature and does not interfere with the pressure assumptions used elsewhere. With this change, when the recycle (tear) stream temperature is increased by 2 °C to represent the disturbance, the loop no longer amplifies the temperature; after convergence, the recycle temperature returns to essentially its original level, showing the added cooler successfully breaks the positive-feedback path and makes operational-mode behavior match the design-mode result.

Because this block is removing heat, the most consistent default Aspen utility is high-pressure steam generation (HPSTEAM), since it represents recovering high-grade heat as exportable steam (and it is closer to the process temperature level than the other default utility choices). The HP steam generation cost rate is taken directly from the Aspen Process Economic Analyzer utility summary for E-201, where it appears as a negative cost per hour because it is treated as steam you can sell. Using the APEA result of –1.79194 USD/h, the annual utility credit is:

$$\Delta C_{\text{utilities}} = (-1.79194 \text{ USD/h}) \times (8000 \text{ h/yr}) = -14,335.52 \text{ USD/yr}$$

So the mitigation provides an annual utility cost decrease of about \$14.3k/yr (credit from exporting HP steam). Aspen also lists an electricity line item that is not clearly tied to a specific unit here, so it is not used in the net-utility comparison for this calculation.

Refer to the figure below for the updated flowsheet showing E-201 placed upstream of the reactor, with the inlet temperature specification (310 °C) and no pressure drop, and the resulting stable loop behavior after applying the +2 °C tear-stream disturbance.

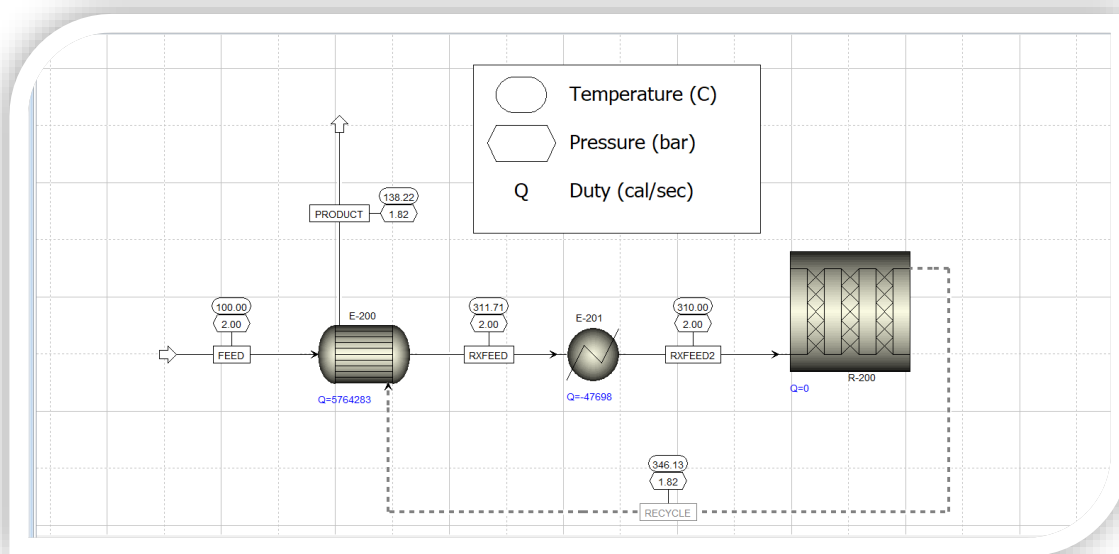


Figure 9: post-solution Aspen Flowsheet

References

Aspen Technology, Inc. *Aspen Plus User Guide, Version 14*. Bedford, MA: Aspen Technology, Inc., 2023.

Caracostas, Michael. *CHE Calculator, Version 13*. Excel-based chemical engineering calculation and dynamic modeling tool. University of Illinois Chicago, Department of Chemical Engineering, 2025.

OpenAI. *ChatGPT* (GPT-based large language model). San Francisco: OpenAI, accessed January 2026. <https://chat.openai.com>.